

This rubric is intended to help standardizing the evaluation of the final assignment (report/paper) of this course, making grading more consistent and transparent for students. Also it will give some proper feedback to students when communicating the grade. And it will let the students know beforehand what aspects they will be evaluated on, possibly for a critical self-evaluation of their own work.

The assignment is a multi-dimensional project to be carried out in groups of 2 to 3 students. As a group, students will first define a flow problem (approval needed), which involves at least two components of heat transfer, chemical reaction, and turbulence. They will then numerically solve this flow problem using the supplied code or a self-written code. With the same group they will finally write it out as a scientific paper following the supplied template.

Rubric Introduction to CFD (4RC30)

Performance Criteria	Weight	Exceptional Quality	Average Quality	Poor Quality
Level of difficulty	5%	The assignment solves one of the asked flow problems, which has a high level of difficulty with a lot self-implementation of the code.	The assignment solves one of the asked flow problems, which has a certain level of difficulty with some self-implementation of the code.	The assignment solves one of the asked flow problems, which is however simple as one of the tutorials or examples introduced during the course.
Analysis of the flow problem	10%	The student well understands and clearly describes the problem studied. • Perfect description of the transport equations, initial and boundary conditions, reasonable assumptions.	The student understands and fairly describes the problem studied. • Most information about the transport equations, initial and boundary conditions, reasonable assumptions.	The student doesn't understand or fails to describe the problem studied. • Missing important information about the transport equations, initial and boundary conditions, reasonable assumptions.
CFD model	25%	The developed model is clearly described and is an exact representation of the studied flow problem.	The developed model is clearly described but shows some minor mistakes.	The developed model is not clearly described and fails to represent the studied flow problem.
Execution of CFD model	20%	• Complete code implementation of the developed model. • Execution of the model/code shows great attention for the convergence, efficiency and accuracy (grid resolution, time step, turbulence model, slip model, differencing schemes, etc.)	• Complete code implementation of the developed model. • Execution of the model/code shows certain attention for the convergence, efficiency and accuracy (grid resolution, time step, turbulence model, slip model, differencing schemes, etc.)	• Incomplete code implementation of the developed model. • Execution of the model/code shows problems concerning the convergence, efficiency and accuracy (grid resolution, time step, turbulence model, slip model, differencing schemes, etc.)
Solutions/Results	25%	The results meet the objective(s) and are explicitly detailed and analyzed.	The results meet the objective(s) and are fairly detailed and analyzed.	The results don't meet the objective(s) and/or are poorly detailed and analyzed.

Figures and Tables	10%	All figures and tables are well created and communicate their purpose clearly.	Most figures and tables are well created and communicate their purpose clearly.	Only some figures and tables are well created and communicate their purpose clearly.
Writing and layout • Grammar & Spelling • Structure • Readability • References	5%	Excellent scientific writing • All grammar and spelling are correct • Structure is clear and complete and shows great care in the layout design • Paper is well readable • Right citations and references	Legible scientific writing • Only a few grammar and spelling errors • Structure is complete but does not show a careful design of the layout • Paper is readable • Right citations and references	Illegible scientific writing • Many grammar and spelling errors • Structure is not complete • Paper is unreadable • Missing citations and references